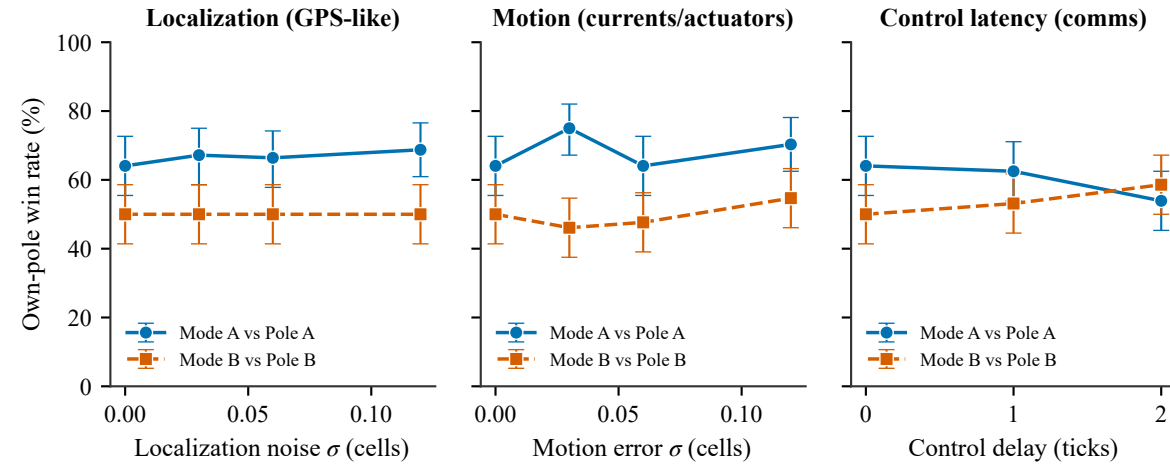




Sim-to-real diagnostic dose-response ($n=128$). Families map to physical deployment: localization $\rightarrow$ GPS/pose uncertainty; motion $\rightarrow$ currents/wind/actuator mismatch; delay $\rightarrow$ communications/control latency. Matched numeric schedule for localization/motion (0.03/0.06/0.12 cells); delay 0/1/2 ticks. Error bars: 95% bootstrap CIs.